Moment of Inertia of a Solid Cone about Its Axis

Stewart, Calculus: Early Transcendentals, Section 15.7, Exercise 46

Problem

Assume that the solid has constant density k. Find the moment of inertia about the z-axis of the solid cone $\sqrt{x^2+y^2} \le z \le h$.

Symbols

- E is the solid cone. Its apex is at the origin and its axis is the z-axis.
- h is the height of the cone. The cone also has radius h at its base, because the surface satisfies $\sqrt{x^2+y^2} = z$.
- k is the constant density, that is, mass per unit volume.
- r is the distance from the point (x, y, z) to the z-axis: $r = \sqrt{x^2+y^2}$. In cylinder coordinates, $x = r\cos\theta$ and $y = r\sin\theta$.
- dV is the volume element. In cylinder coordinates, $dV = r\,dz\,dr\,d\theta$. The extra factor r is the Jacobian.
- I_z is the moment of inertia about the z-axis: $I_z = \iiint_E (x^2+y^2)\,\rho\,dV$. The factor x^2+y^2 is the square of the distance from the point to the axis of rotation.
- m is the total mass, $m = \iiint_E \rho\,dV$.

The moment of inertia measures how strongly the solid resists rotation about the axis. Mass far from the axis counts more, because the distance is squared.

Part (a): moment of inertia about the z-axis

Step 1: describe the solid

The two inequalities define the upper nappe of a cone:

$$0 \le \sqrt{x^2+y^2} \le z \le h. \tag{1}$$

In cylinder coordinates this becomes

$$E = \{(r,\theta,z) : 0 \le \theta \le 2\pi,\ 0 \le z \le h,\ 0 \le r \le z\}. \tag{2}$$

At height z the cross-section is a disk of radius z. The disk grows from a point at the apex $(0, 0, 0)$ to radius h at the base $z = h$.

Figure 1 shows the cone with one radial slice. Figure 2 shows the slice in the (r, z)-plane.

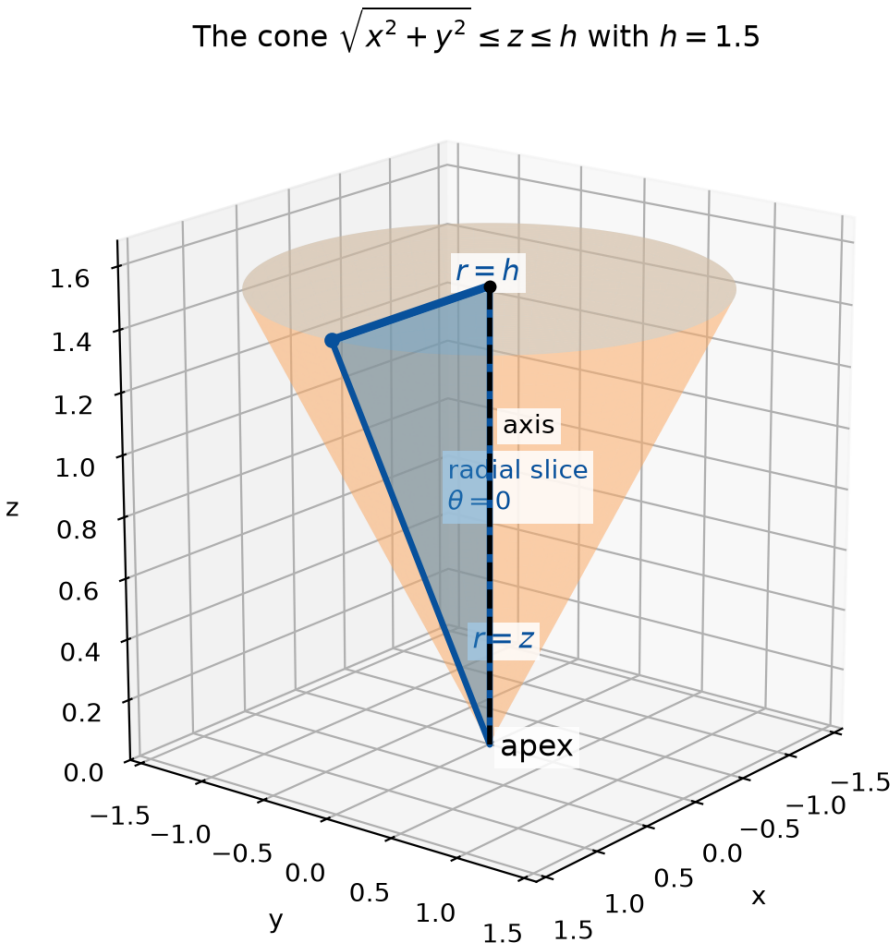


Figure 1: The cone $\sqrt{x^2 + y^2} \le z \le h$ with $h = 1.5$. The orange surface is the cone, the light blue disk is the base, and the blue triangle is the radial slice at $\theta = 0$. The dashed line is the z-axis.

Step 2: set up the integral

The distance from (x, y, z) to the z-axis is r. Therefore $x^2 + y^2 = r^2$ and

$$I_z = \iiint_E (x^2 + y^2)\, k\, dV = k \int_0^{2\pi} \int_0^h \int_0^z r^2 \cdot r\, dr\, dz\, d\theta. \tag{3}$$

The order is legal. The inner limit $r \le z$ uses only z. The middle limit $z \le h$ is constant. The outer limit for θ is constant.

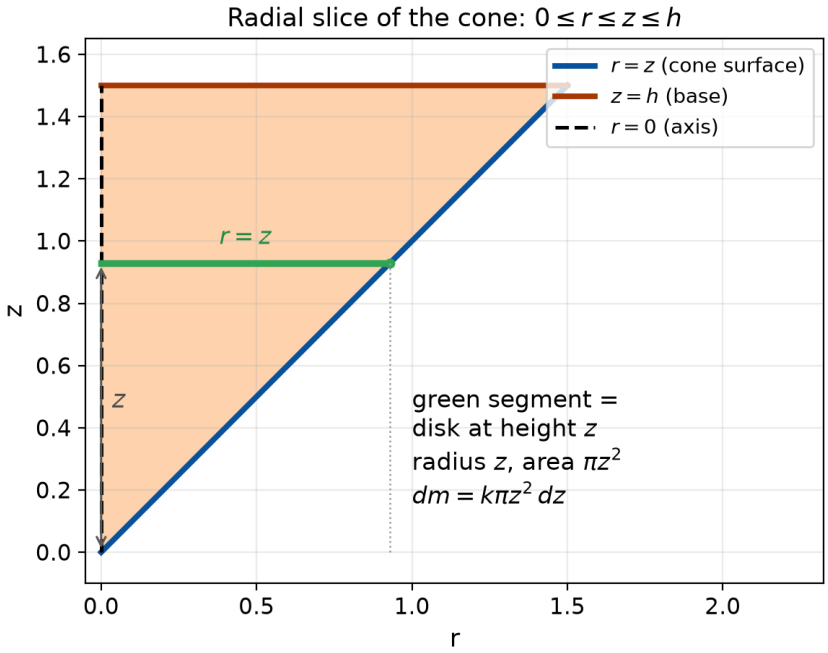


Figure 2: The radial slice: the triangle $0 \le r \le z \le h$. The green segment is the disk at height z. Its radius is z, so its area is πz^2 and its mass is $k\pi z^2\, dz$.

Step 3: integrate over r

$$\int_0^z r^3\, dr = \frac{z^4}{4}. \tag{4}$$

So

$$I_z = k \int_0^{2\pi} \int_0^h \frac{z^4}{4}\, dz\, d\theta. \tag{5}$$

Step 4: integrate over z and θ

$$\int_0^h \frac{z^4}{4}\, dz = \frac{h^5}{20}, \tag{6}$$

$$I_z = k \int_0^{2\pi} \frac{h^5}{20}\, d\theta = k \cdot 2\pi \cdot \frac{h^5}{20} = \frac{\pi k h^5}{10}. \tag{7}$$

Thus

$$I_z = \frac{\pi}{10}\, k\, h^5. \tag{8}$$

This is a constant, $\pi/10$, times k and h^5, as the dimensions require.

Step 5: state the mass

The same limits with integrand $k\, r$ give the mass:

$$m = k \int_0^{2\pi} \int_0^h \int_0^z r\, dr\, dz\, d\theta = k \cdot 2\pi \cdot \frac{h^3}{3} = \frac{\pi k h^3}{3}. \tag{9}$$

This is the standard volume $\pi h^3/3$ of a cone of radius h and height h, multiplied by the density. Elimination of k gives the compact form

$$I_z = \frac{\pi k h^5}{10} = \frac{3}{10}\left(\frac{\pi k h^3}{3}\right) h^2 = \frac{3}{10}\, m\, h^2. \tag{10}$$

Step 6: checks

Check by disks (rotational symmetry). Every horizontal slice is a disk of radius z and mass $dm = k\pi z^2\, dz$. For a uniform disk, the moment of inertia about its own center is $\frac{1}{2}(\text{mass})(\text{radius})^2$. Because the disk is centered on the z-axis, this is also its moment about the z-axis. Hence

$$I_z = \int_0^h \frac{1}{2}\, z^2\, dm = \int_0^h \frac{1}{2}\, z^2\, k\pi z^2\, dz = \frac{k\pi}{2}\int_0^h z^4\, dz = \frac{\pi k h^5}{10}. \tag{11}$$

This is the same value from a different argument. The symmetry of the solid about the z-axis is what makes the disk formula legal.

Symmetry of the two transverse moments. The cone is unchanged by a rotation about the z-axis, so $I_x = I_y$. A direct computation gives

$$I_x = k\iiint_E (y^2 + z^2)\, dV = \frac{\pi k h^5}{4}. \tag{12}$$

For any solid of revolution about the z-axis, each thin disk satisfies $I_x = \frac{1}{2} I_z + z^2\, dm$. Integration gives

$$I_x = \frac{1}{2}\cdot\frac{\pi k h^5}{10} + k\pi\int_0^h z^4\, dz = \frac{\pi k h^5}{20} + \frac{\pi k h^5}{5} = \frac{\pi k h^5}{4}, \tag{13}$$

which agrees. This also matches the standard table entry $I_x = \frac{3}{20} m(4h^2 + h^2) = \frac{3}{4} m h^2$ for a cone with $R = h$.

Dimensions. $[k] = M/L^3$ and $[h^5] = L^5$, so $[kh^5] = ML^2$, which is the correct unit for a moment of inertia. The mass form $I_z = \frac{3}{10} m h^2$ has the same unit.

Numerical confirmation

Two numerical methods test the result with $h = 1.7$ and $k = 2.5$. The first method is a midpoint Riemann sum in cylinder coordinates on the triangle $0 \le r \le z \le h$ with 2400^2 cells. The second method is a Monte Carlo average with 4×10^6 points in the box $[-h, h] \times [-h, h] \times [0, h]$, using the Cartesian test $x^2 + y^2 \le z^2$. Table 1 lists the values and the relative errors.

Answer

$$\boxed{I_z = \frac{\pi}{10}\, k\, h^5, \qquad m = \frac{\pi}{3}\, k\, h^3, \qquad I_z = \frac{3}{10}\, m\, h^2} \tag{14}$$

Quantity	Exact value	Riemann sum	Monte Carlo
m	12.862204	12.870244	12.874271
I_z	11.151531	11.163149	11.154344
$I_z/(mh^2)$	0.300000	0.300125	0.299794

Table 1: Numerical checks for Exercise 46 with $h = 1.7$ and $k = 2.5$. The relative error is at most 9.4×10^{-4} for the mass and 1.1×10^{-3} for I_z.

How to solve problems of this kind

1. Identify the axis of rotation first. The integrand of I is the square of the distance to that axis: $x^2 + y^2$ for the z-axis, $y^2 + z^2$ for the x-axis.
2. For a solid with circular symmetry, use cylinder coordinates. Then $x^2 + y^2 = r^2$ and $dV = r\,dz\,dr\,d\theta$.
3. Write the solid as inequalities in (r, θ, z). For this cone the surface is $r = z$, so $0 \le r \le z$.
4. Find the mass with the same limits. Only the integrand changes, from r^3k to rk.
5. Use a slice check. A thin disk has $I = \frac{1}{2}ma^2$ about its center. This reproduces the answer without a triple integral and tests the symmetry.
6. Check the dimensions, then compare with the standard table value for a cone. Agreement with a known formula is strong evidence.
7. Confirm with one numerical method in a different coordinate system. A Cartesian Monte Carlo test catches errors in the Jacobian.

Summary of results

Quantity	Symbol	Value
height and base radius	h	arbitrary
density	k	arbitrary
mass	m	$\pi kh^3/3$
moment about z-axis	I_z	$\pi kh^5/10$
moment in mass form	I_z	$\frac{3}{10}mh^2$
transverse moment	$I_x = I_y$	$\pi kh^5/4 = \frac{3}{4}mh^2$

Table 2: All exact values for Exercise 46.